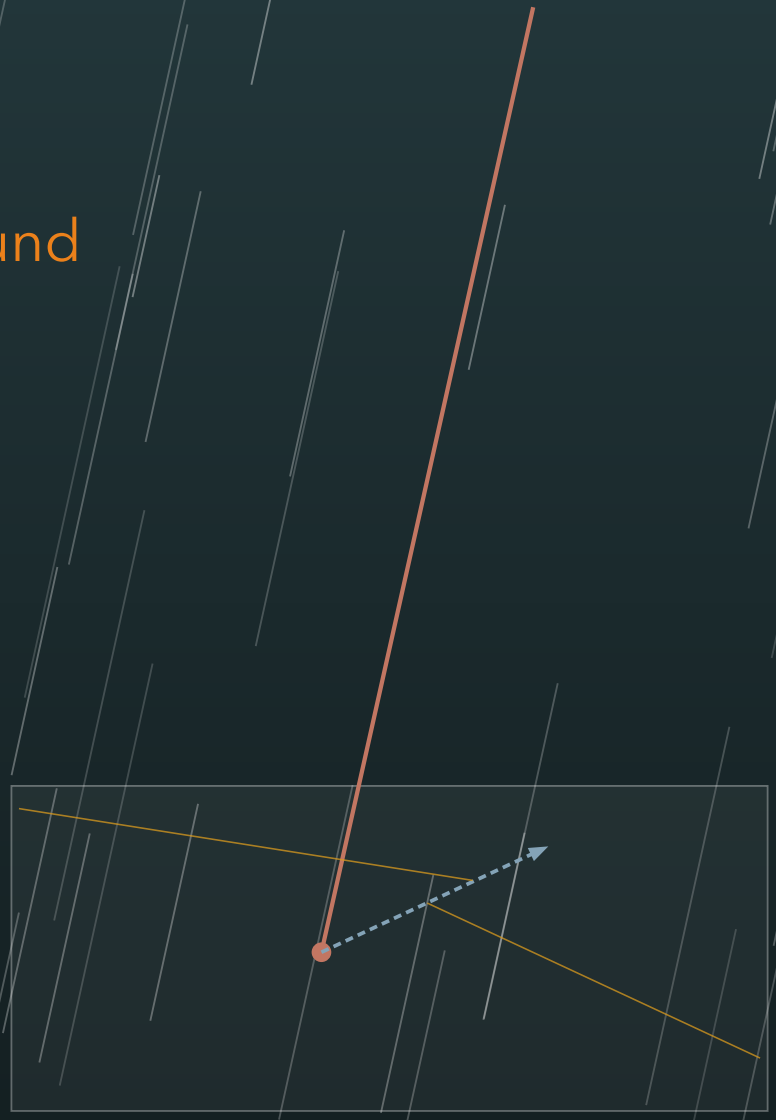
Muon Lifetime

Timing particles that should never reach the ground

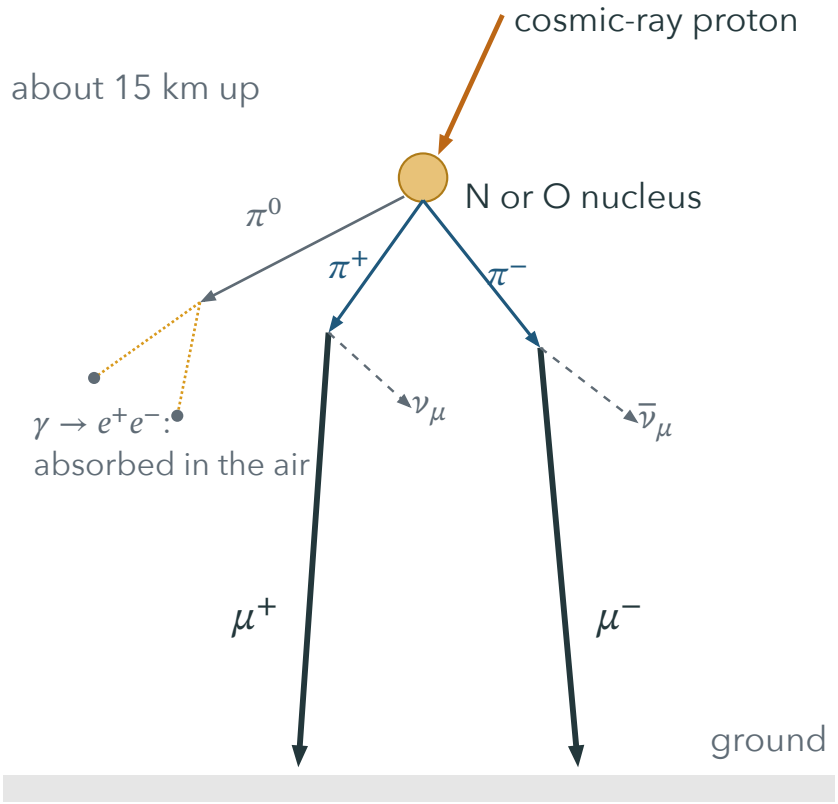
Made 15 km up. Living 2.2 μs: at the speed of light, 660 m. Yet one crosses every square centimetre of you each minute.

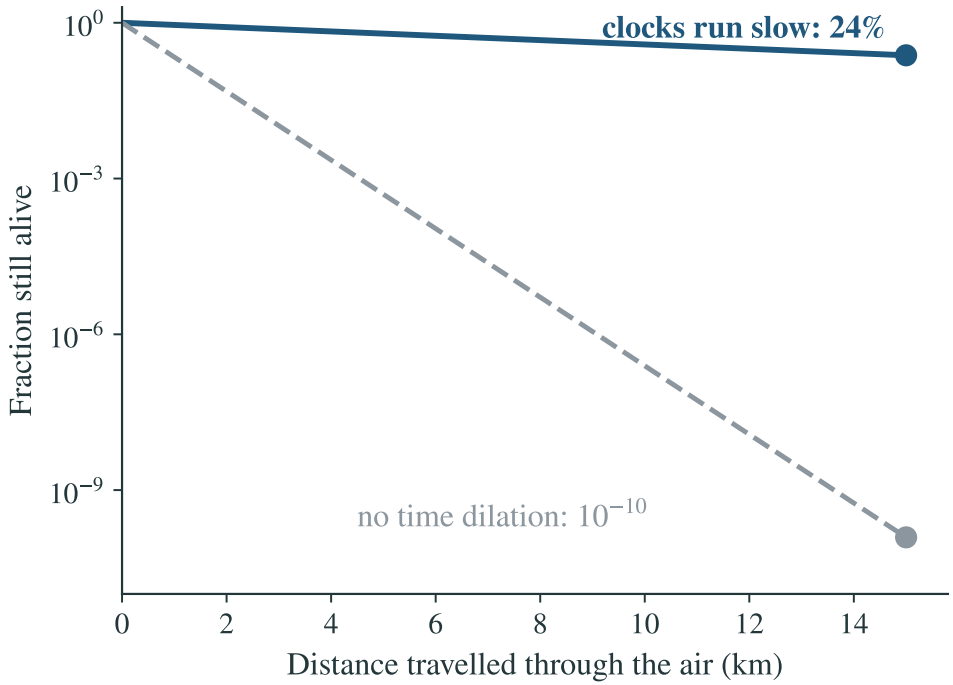
How?



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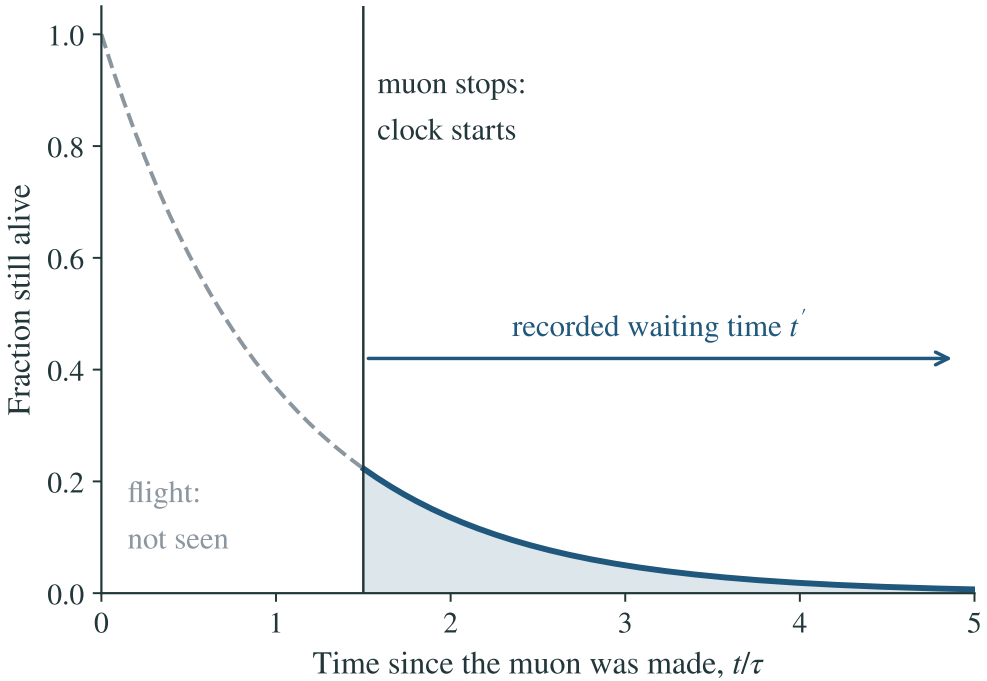
Muons reach the ground because their clocks run slow





Question: can a stable lifetime be extracted when the detector adds signals of its own?
Answer: $\tau = (2.17 \pm 0.04\ (\text{stat}) \pm 0.04\ (\text{sys}))\ \mu\text{s}$, consistent with 2.197 µs

Timing a stopped muon measures its full lifetime



No memory: the same chance to decay in every instant,

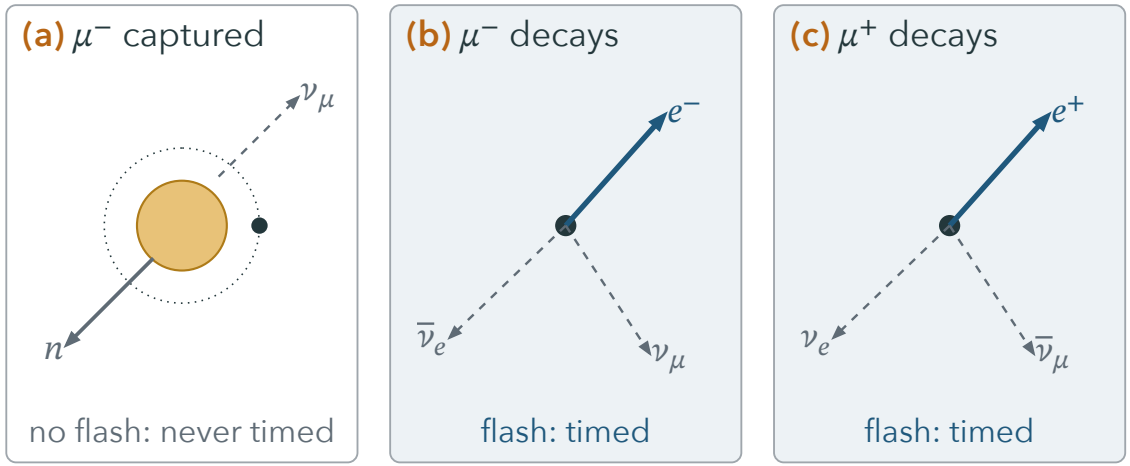
$$N(t) = N_0 e^{-t/\tau}$$

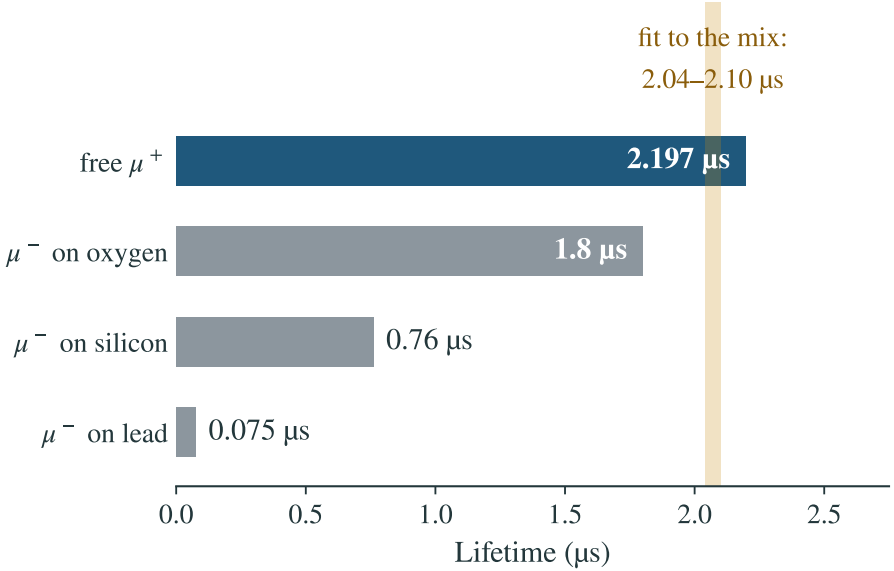
so the unknown flight time cancels:

$$\frac{N(t_a + t')}{N(t_a)} = e^{-t'/\tau}$$

A fixed delay in starting the clock cancels the same way: it changes the amplitude, not τ

Capture can only shorten the measured lifetime

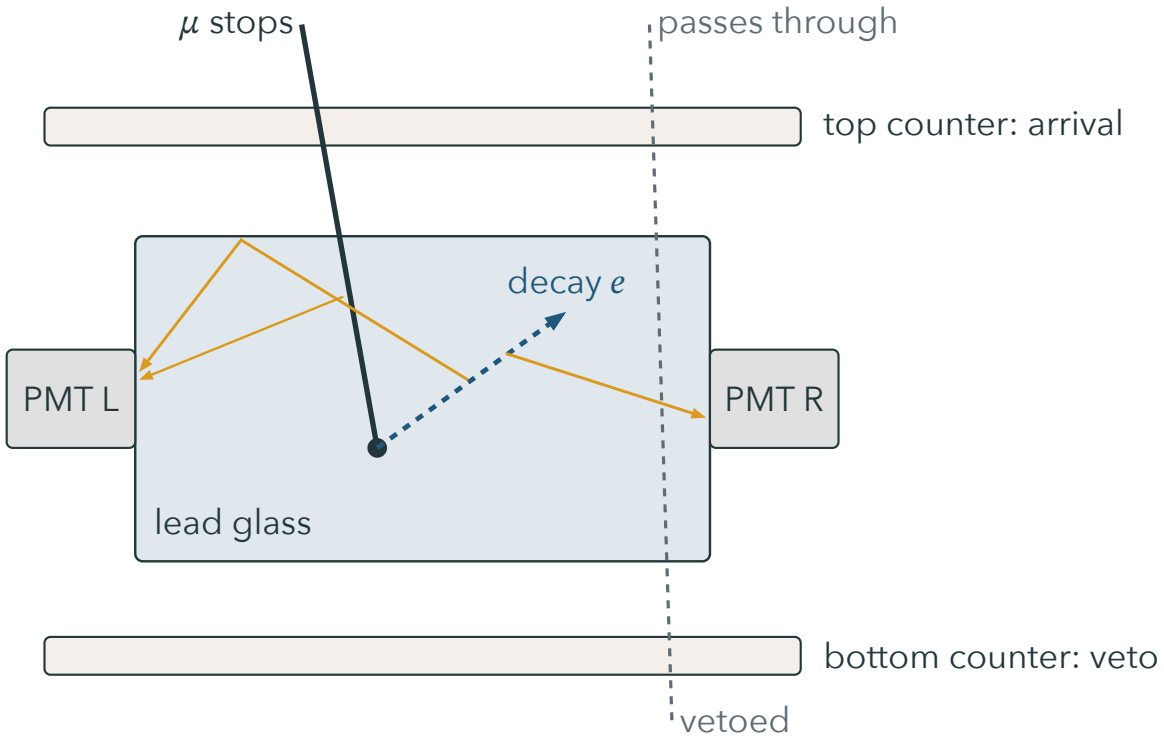




A captured μ^- never makes a decay electron. Capture and decay rates add, $1/\tau_j = 1/\tau_\mu + \Lambda_j$, so the fit can only come out shorter.

Expected fit assumes 45% PbO and 46% SiO_2 by mass, $\mu^+/\mu^- = 1.27$, and capture shared by the Fermi–Teller Z law; bound lifetimes from [1] T. Suzuki, D. F. Measday, and J. P. Roalsvig, Total nuclear capture rates for negative muons, Physical Review C 35, 2212 (1987)

The glass sees a muon stop, then its electron flash



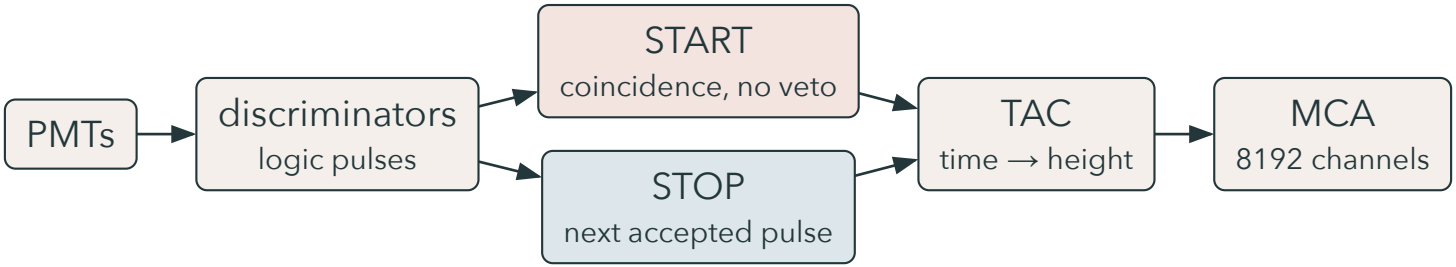
Lead glass gives **Cherenkov light** only above $\beta = 1/n$:

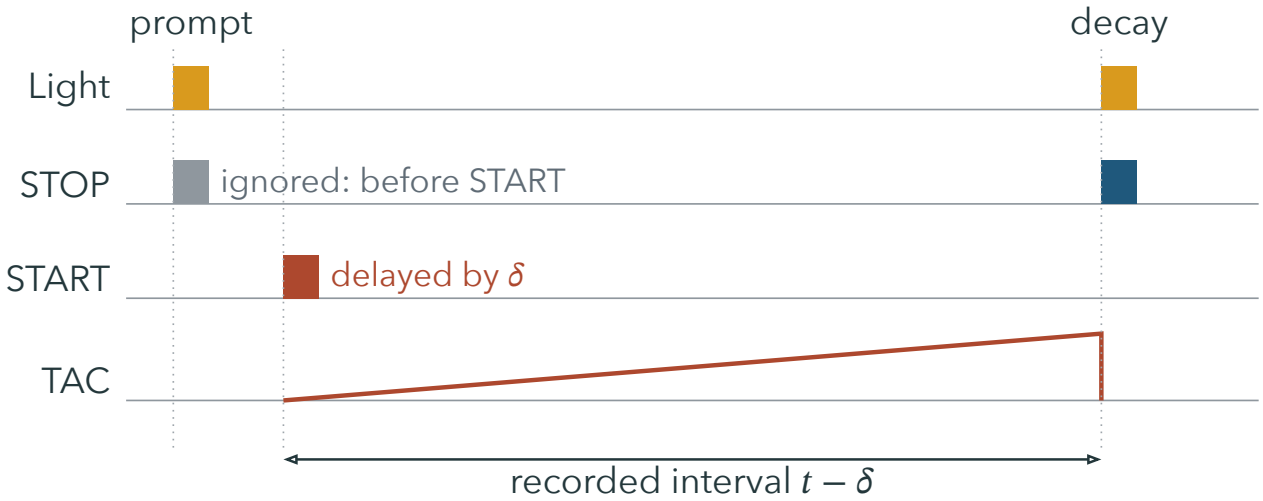
electron	0.14 MeV
muon	29 MeV

The stopping muon goes dark; its decay electron, up to 53 MeV, always flashes[2]

[2] B. Dickson, Determination of Mean Muon Lifetime, (2014)

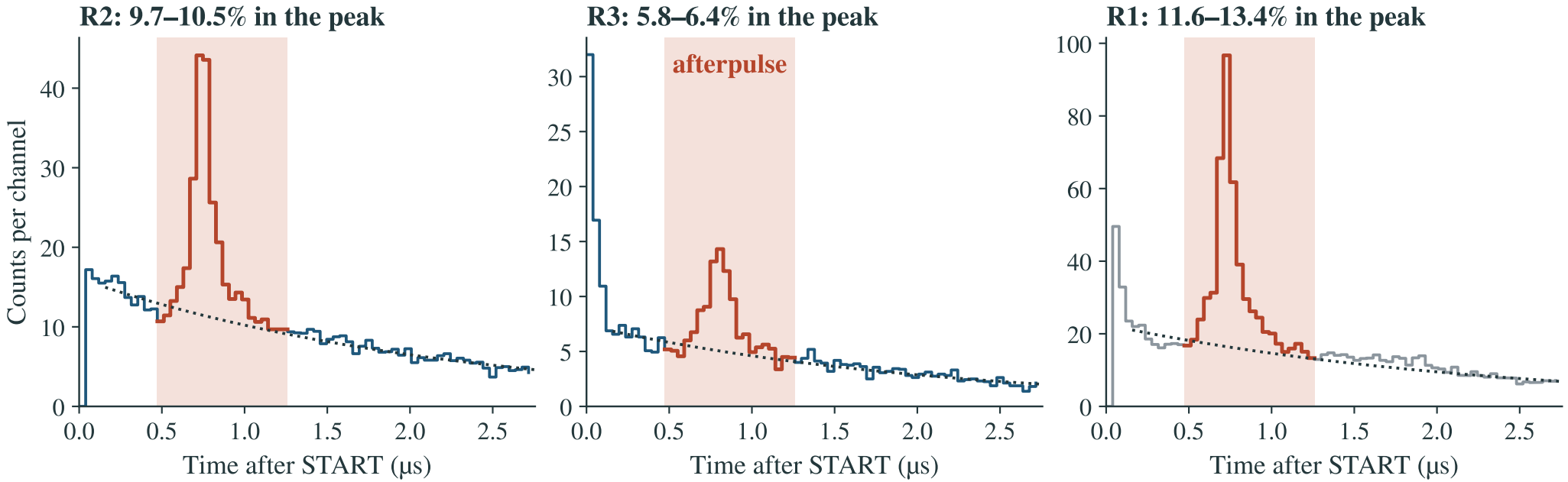
Two flashes become one time interval





The muon's own prompt flash would stop the clock at once, so the START waits by δ; the fixed delay cancels. Three acquisitions, R1–R3, used different settings.

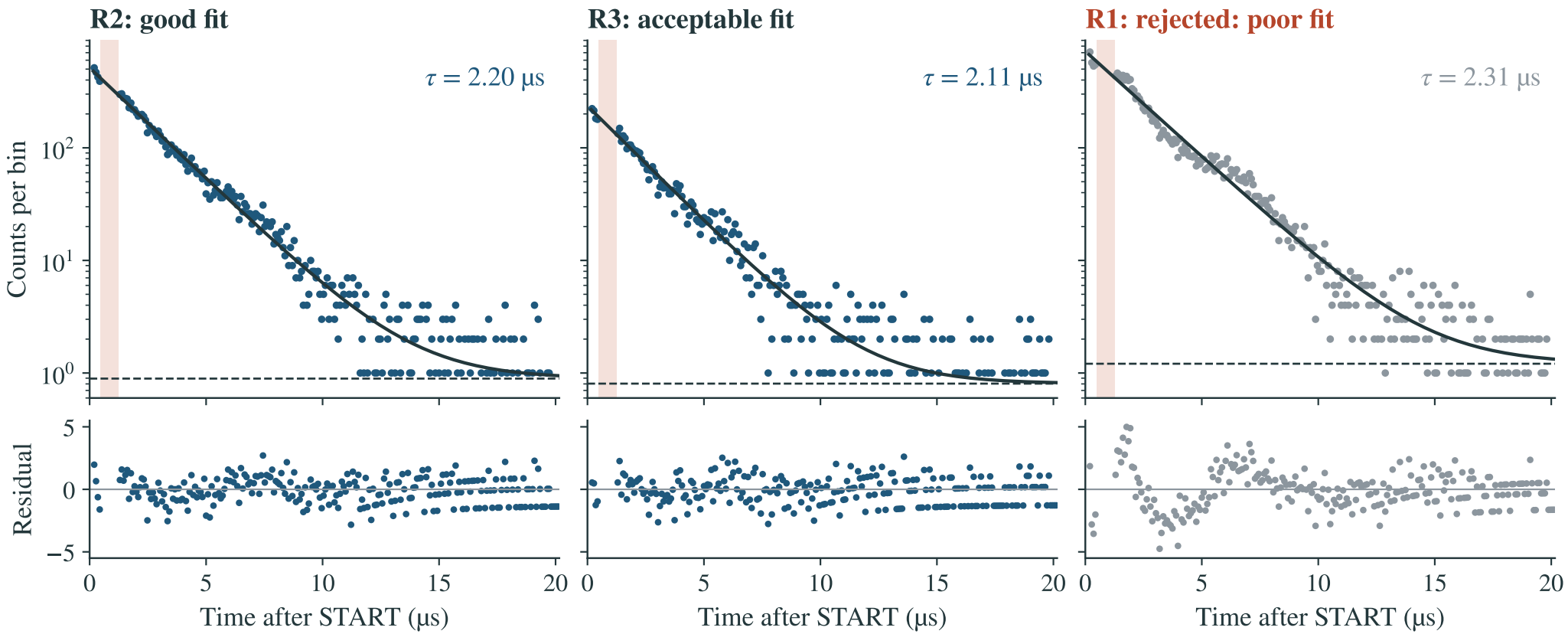
The photomultiplier adds a peak no muon can make



Same delay in all three, despite changed voltage, threshold and coincidence: an ion-feedback afterpulse[3], excluded from the fit

[3] Hamamatsu Photonics, Photomultiplier Tubes and Related Products: Construction and Operating Characteristics, (n.d.)

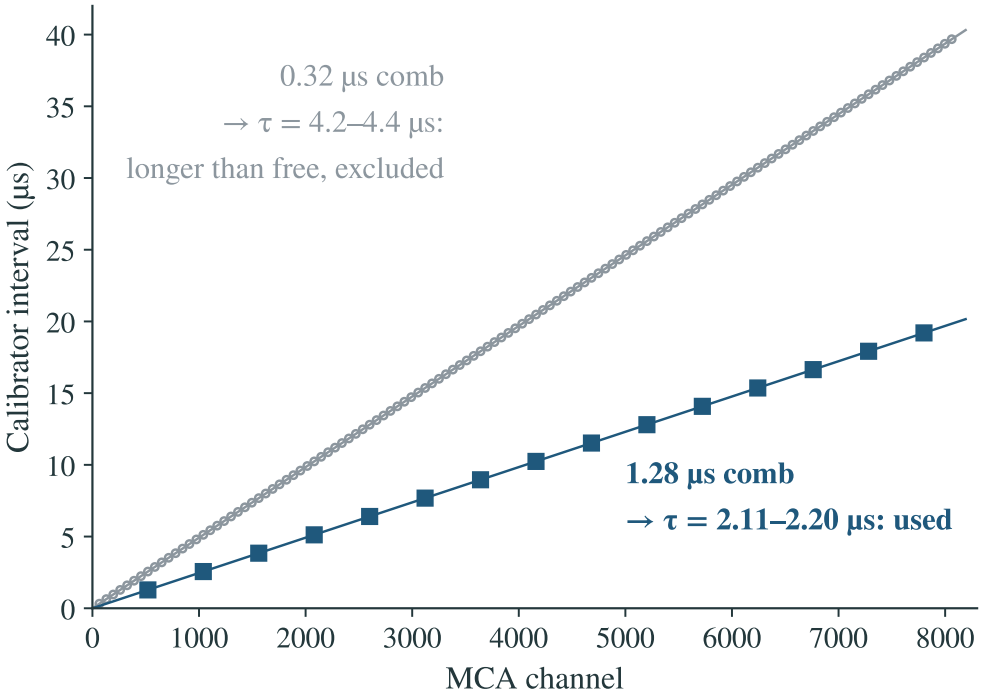
Fitting around the peak: two acquisitions fit, one does not



One exponential plus a flat background, fitted on both sides of the peak

Deviance for 241 degrees of freedom: R2 274.3, R3 292.4, R1 614.9. Poisson likelihood on 32-channel bins, channels 64–192 and 512–8192; signed deviance residuals. Times on the 1.28 µs calibration.

The time scale rests on one unverified calibration

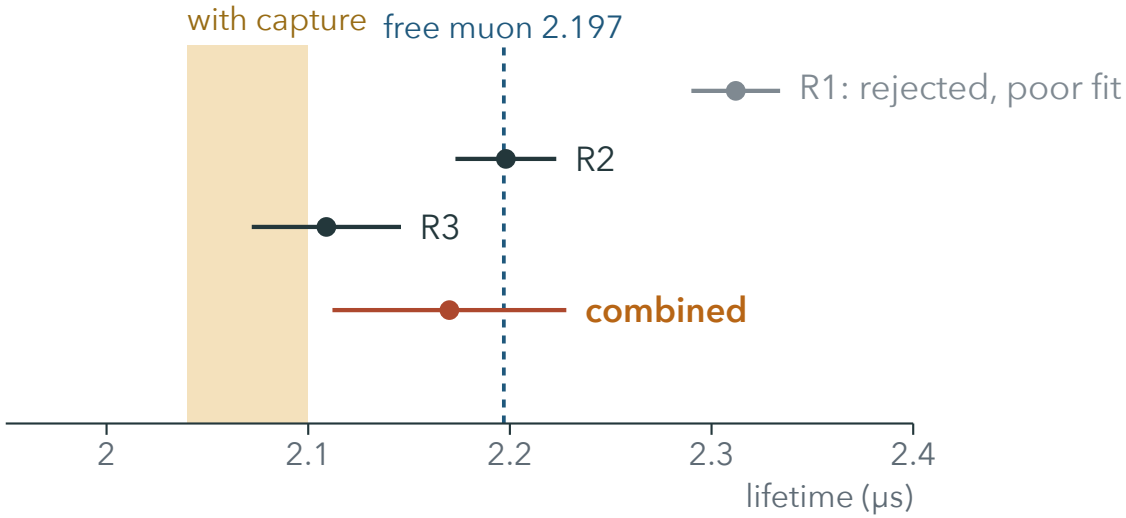


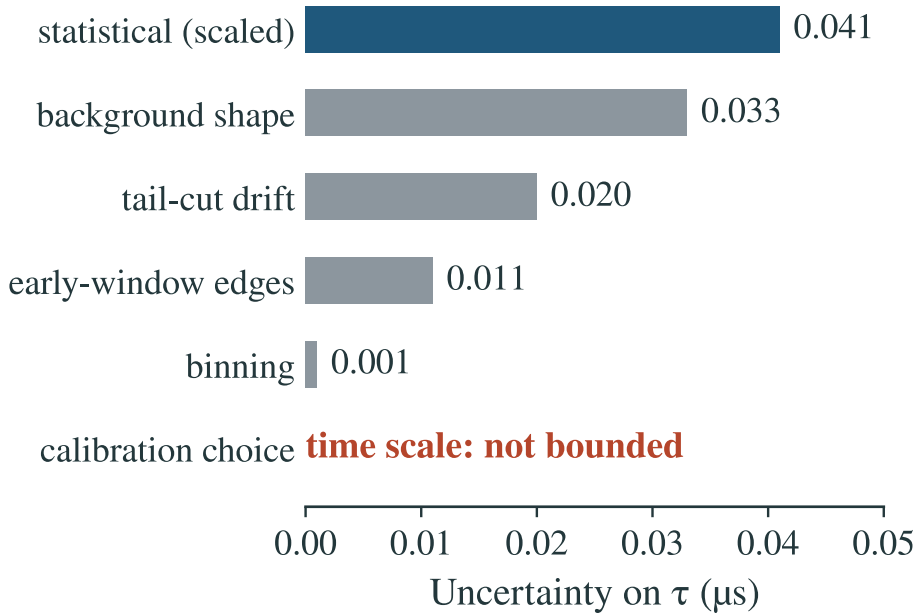
The two combs differ by a factor of **two**, not the expected four.

Capture can only **shorten** τ, so the 0.32 µs scale cannot apply.

The 1.28 µs scale was never recorded under the decay settings: **not bounded**.

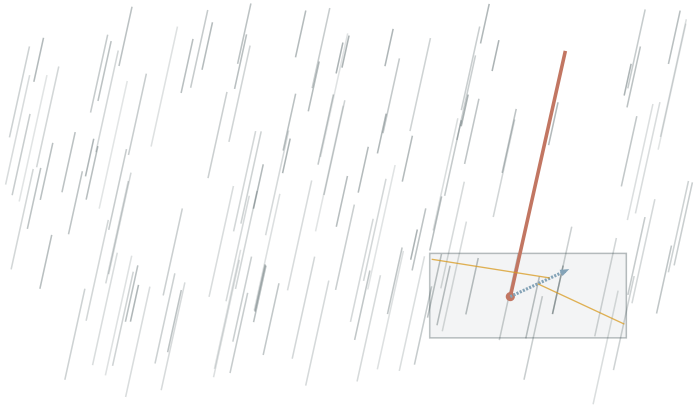
Combined: $\tau = (2.17 \pm 0.04\ (\text{stat}) \pm 0.04\ (\text{sys}))$ μs





R2 and R3 disagree ($\chi^2 = 3.9$), so their spread scales the uncertainty by $S = 2.0$. Consistent with 2.197 μs; the capture shift is not resolved.

A lifetime consistent with 2.197 µs, limited by the detector



How do they reach us?	**Their clocks run slow**: about 24% survive 15 km
What lifetime?	$\tau = (2.17 \pm 0.04\ (\text{stat}) \pm 0.04\ (\text{sys}))$ µs, consistent with 2.197 µs
What limits it?	an afterpulse in every spectrum; settings that change the result; **an unverified time scale**

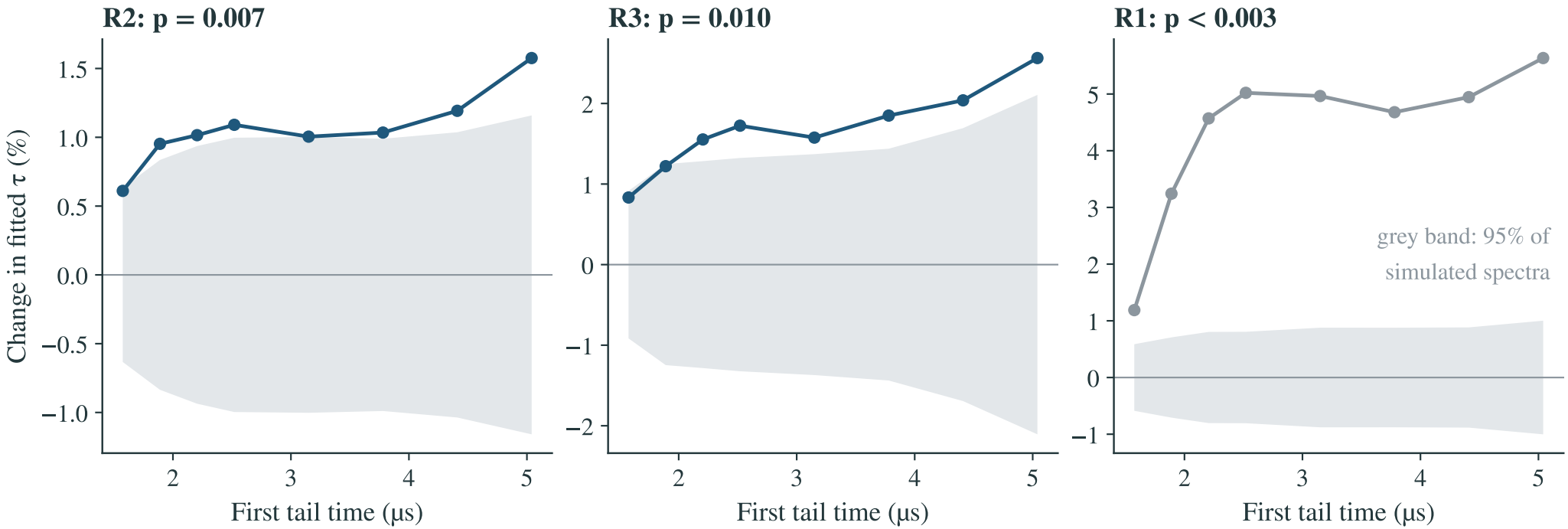
Next

- calibrate under each run's settings
- oscilloscope test of the afterpulse
- measure the glass composition

Uncertainty contributions by acquisition

Source (μs)	R2	R3	Method
Counting statistics	0.025	0.037	profile likelihood
Background shape	0.037	0.025	flat vs. sloping
Tail-cut drift	0.017	0.027	cuts 512–2048
Early-window edges	0.013	0.008	edge scans
Binning	0.001	0.001	widths 16–128
Accidental STOPs	≤ 0.003	≤ 0.006	one-sided bound
Total systematic	0.043	0.038	quadrature
Calibration choice	n/a	n/a	not bounded

The tail-cut drift



Starting the tail fit later lengthens the decay in all three, more than chance allows. Cause not established; half its range enters the systematic uncertainty.

References

[1] T. Suzuki, D. F. Measday, and J. P. Roalsvig, Total nuclear capture rates for negative muons, Physical Review C **35**, 2212 (1987).

[2] B. Dickson, Determination of Mean Muon Lifetime, (2014).

[3] Hamamatsu Photonics, Photomultiplier Tubes and Related Products: Construction and Operating Characteristics, (n.d.).